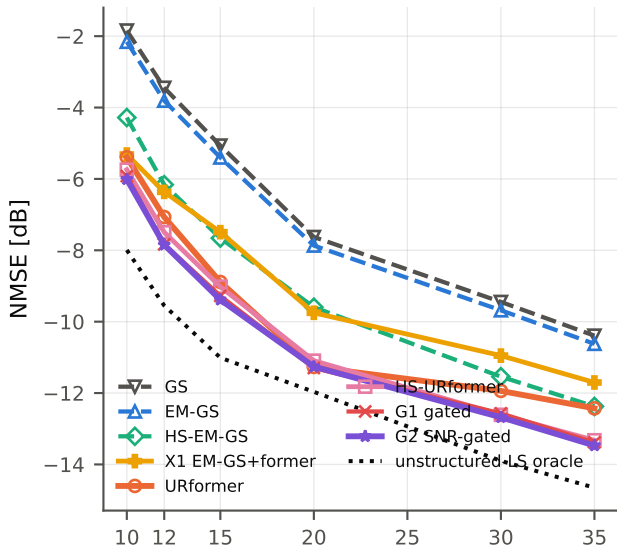
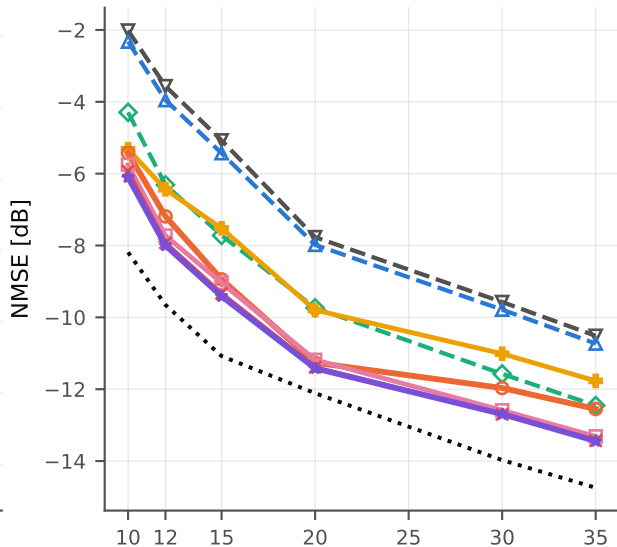


NMSE vs pilots — all methods (SNR = 5 dB)

a. per-trial MEDIAN (project primary)



b. ratio-of-sums (paper convention)



Number of pilots P

Number of pilots P

K=3 users, M=32 atomic antennas, RSR=10 dB (repo convention; 5.23 dB in the paper's multi-user convention), $L_k \sim U\{3..7\}$ paths, 400 trials per point. Classical arms dashed, learned arms solid. All learned arms trained at P=20 on 80k samples (URformer/HS-URformer/G1/G2) or on cached EM-GS (X1). PILOT SWEEP IS A GENERALIZATION TEST: every learned arm was trained at P=20 and is evaluated unchanged at other P. Nothing was retrained.